Yeast Fermentation

Teacher guide | Gas-syringe lab | Biology 2e, Chapter 7

1. Organize a full comparison set

Question: How does the available sugar affect gas accumulation in a yeast culture? This download is specifically for gas-syringe lab. Use the same method for all flasks in a pooled comparison.

Equipment and measurement

Matched flasks, fitted one-hole bungs, delivery tubing and suitable graduated gas syringes, stands and clamps. Also use timers, thermometer, transfer tools, stocks and eye protection.

Gas from the flask moves a freely sliding gas-syringe plunger. Read the scale to measure collected gas volume in mL. The virtual lab uses this quantitative apparatus.

One set = 9 independent flasks: N1-N3 (no added sugar), G1-G3 (glucose), S1-S3 (sucrose). Each flask is a fresh mixture. Multiple readings from one flask are not additional independent trials.

Students	Example allocation of the 9 flasks
18: nine pairs	Each pair prepares 1.
12: six pairs	Three pairs prepare 2; three pairs prepare 1.
10: five pairs	Four pairs prepare 2; one pair prepares 1.
9: three pairs and one trio	Each team prepares 2; the teacher prepares the ninth. Or assign the trio 3.

Assign one or two flask IDs per pair and record ownership on page 4. Roles: mix/prepare, measure/time, record/share. Pairs can combine roles. Each flask needs its own start time and elapsed-time record.

Demo: distribute observation and analysis roles while the teacher operates equipment. If using just one flask per treatment, report n = 1; do not call three time readings three trials. Share matched independent class trials to complete a set.

Allow about 10-15 minutes to plan/prepare, 30 minutes to observe, and 15-20 minutes to pool/analyze. Use a second lesson for analysis if needed. Pilot the equipment and small-flask volumes before class.

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2. Stocks, flask sizes, and supply planning

Choose one flask size and recipe per pooled set. These are proposed pilot volumes, not a promise of measurable gas in one lesson. Keep all treatments at the same measured temperature; a teacher-managed 30 C water bath is one possible condition.

Flask capacity	Treatment stock*	Yeast suspension	Final liquid
50 mL	10 mL	10 mL	20 mL
100 mL	20 mL	20 mL	40 mL
Custom: ____ mL	____ mL	____ mL	____ mL

*Plain water for N; glucose stock for G; sucrose stock for S. Examples leave 60% headspace for gas/foam. Lower the fill if the pilot requires it; do not fill a 50 mL flask with 50 mL culture.

Optional stock recipe

For each sugar: dissolve 80 g glucose OR sucrose and bring to a final volume of 1,000 mL with water (8% mass/volume). Disperse 20 g dry baker's yeast and bring to 1,000 mL (2% suspension). Keep plain water separately. Equal volumes of sugar stock and yeast give 4% sugar and 1% yeast in the flask. N has no added sugar.

These match the virtual mass concentrations, not its 50 mL culture volume. Record the glucose product (including any dextrose monohydrate). Equal sugar mass is not equal molarity. Teachers may substitute a piloted recipe.

Prepare labeled sugar stocks ahead. Prepare yeast close to use following package hydration guidance. Do not add sugar to shared yeast stock. Gently resuspend before EVERY aliquot because yeast settles. Use separate labeled transfer tools and equilibrate all stocks to the chosen test temperature.

For one 9-flask set	50 mL flasks	100 mL flasks
Plain water / glucose / sucrose (each)	30 mL	60 mL
Yeast suspension (all 9)	90 mL	180 mL
Sets from 1 L of EACH stock*	11	5

*Before losses; yeast is limiting. For B sets and V mL of each stock per flask: each treatment stock = 3 x B x V; yeast = 9 x B x V. Allow extra for spills, pilot and optional controls.

Chosen sugar stock: ____ g / ____ mL. Yeast: ____ g / ____ mL. Final sugar concentration: ____ %. Test temperature: ____ C.

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3. Set up, measure, and interpret

Support the gas syringe horizontally as its instructions specify, with the plunger free to move. Connect the flask to the syringe using a fitted bung and open tubing; the flask end stays above the culture. Check joints for leaks. A needle-free medical syringe may have too much friction; use suitable gas-collection equipment.

Label assigned flasks; dispense treatment stock first. Gently resuspend yeast stock immediately before each aliquot. Add yeast last, mix consistently, promptly connect the collector and start that flask's timer. Record the initial reading and any connection delay.

Read at 0, 5, 10, 15, 20, 25 and 30 elapsed minutes, or a shared teacher-selected schedule. Read the syringe scale at eye level at the manufacturer-specified reference point. Record initial volume V0 and each raw volume Vt. Calculate net collected volume Vt minus V0. Do not push or pull the plunger during a run.

Record V0 before the first timed reading. Retain raw Vt readings; calculate net volume = Vt - V0. Flag an unexpected decrease rather than silently replacing it. M = missing; L = suspected leak. Record an observed zero as 0, but never replace missing data with zero.

Pooling and interpretation

Subtract each flask's own V0 before pooling. Average comparable net volumes and report actual n at each time. Keep raw readings and flag leaks, missing data and any exclusions; do not silently discard unusual results.

Plunger friction, leaks, dead space, dissolved CO_2 in the culture and temperature changes can affect results. This directly measures collected volume and avoids extra collection-water loss, but it is not error-free or a measurement of all CO_2 produced.

Keep recipes, temperature, flask size, apparatus and units matched. Keep virtual and physical results separate even if both use mL. For an interval rate, divide the change in net collected volume by elapsed minutes (mL/min). A late interval rate and total gas accumulated are different quantities.

Do not require a winning sugar. Gas accumulation supports a claim about yeast activity; it does not measure ATP yield or establish oxygen absence. The no-added-sugar control can show background activity. An optional sugar-without-yeast control helps check physical expansion.

Practical checks

Wear eye protection; check glassware and follow school procedures. Do not taste cultures. Teacher manages warm water, foaming, cleanup and equipment capacity. Never block the tubing, restrain the moving plunger or cap a gas-producing culture. Stop before the syringe reaches its capacity. Disconnect before resetting or dismantling; the teacher checks apparatus suitability.

If readings do not change: check yeast freshness, temperature, settling, leaks and collector performance. For balloons, inflation threshold may hide small gas amounts. Keep observations, allow more time if practical, and document problems.

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4. Assignment roster and sharing

Class/date: _____ Set ID: _____ Students: ____

Method: Gas-syringe lab | Unit: mL

Flask: ____ mL Treatment stock: ____ mL Yeast: ____ mL

Temperature: ____ C Reading interval: ____ min End time: ____ min

Flask ID	Treatment	Pair/team or demo	Start time
N1	No added sugar		
N2	No added sugar		
N3	No added sugar		
G1	Glucose		
G2	Glucose		
G3	Glucose		
S1	Sucrose		
S2	Sucrose		
S3	Sucrose		

Print worksheet pages 1-2 for each pair/team and pages 3-4 for each student or pooled group. Page 2 records up to two flasks; copy it for additional flasks. Page 3 combines all nine.

Report actual n if fewer than three valid independent trials remain. Keep flagged originals if a fresh replacement is made. Use the classroom worksheet or a separate spreadsheet. The virtual lab stays on a gas syringe; never enter balloon cm into its mL fields.

Method and reading connections

Science World: Yeast-Inflated Balloons https://www.scienceworld.ca/resource/yeast-inflated-balloons/ Science Buddies: An Aerobic Exercise (inverted-cylinder collection) https://www.sciencebuddies.org/science-fair-projects/project-ideas/MicroBio_p009/microbiology/yeast-metabolism-aerobic-anaerobic OpenStax Biology 2e, 7.5: Metabolism without Oxygen https://openstax.org/books/biology-2e/pages/7-5-metabolism-without-oxygen

The small-flask recipes and volumes are proposed pilot settings, not validated performance claims. The virtual apparatus measures collected gas; it does not reproduce every physical measurement error.